LESSON 9.1
ADDING AND SUBTRACTING LINEAR EXPRESSIONS WITH INTEGER COEFFICIENTS

LESSON INTRODUCTION

MA.7.AR.1.1 Apply properties of operations to add and subtract linear expressions with rational coefficients.

LEARNING TARGETS

- I can use algebra tiles to add and subtract linear expressions.

BENCHMARK COHERENCE

BUILDING ON

MA.6.AR.1.4 Apply the properties of operations to generate equivalent algebraic expressions with integer coefficients.

WORKING TOWARD

MA.8.AR.1.2 Apply properties of operations to multiply two linear expressions with rational coefficients.

ESSENTIAL QUESTIONS

- How can you represent algebraic expressions using algebra tiles?
- How do you determine whether terms are like or unlike?
- How could you use zero pairs when simplifying expressions?

LESSON NARRATIVE

Students simplify algebraic expressions using algebra tiles by combining like terms and eliminating zero pairs. Students have prior experience using integer chips to add and subtract rational numbers. They will connect this knowledge to algebra tiles and combining like terms to write equivalent algebraic expressions.

COMPONENT SUMMARY

9.1.1 WARM-UP *(~5 MIN.)*

Students learn about engineers as a Career Profile (video)

9.1.2 REFRESHER *(10 MIN.)*

Exploring zero pairs using algebra tiles

9.1.3 GUIDED INSTRUCTION *(5 MIN.)*

Introduction to academic vocabulary related to algebraic expressions

9.1.4 EXPLORATION *(5-10 MIN.)*

Applying understanding of zero pairs and like terms with algebra tiles to algebraic expressions

9.1.5 GUIDED INSTRUCTION *(10 MIN.)*

Model algebraic expressions using algebra tiles, identify like terms, eliminate zero pairs, and write in simplest form

9.1.6 GUIDED PRACTICE *(5 MIN.)*

Practice adding and subtracting expressions with algebra tiles and writing expressions in simplest form

9.1.7 YOUR TURN! *(5 MIN.)*

Apply understanding of combining like terms to write equivalent algebraic expressions

9.1.8 WRAP-UP *(~5 MIN.)*

Demonstrate understanding of the key vocabulary from the lesson

RESOURCES

GLOSSARY

Key Terms:

- Zero pair
- Term
- Expression
- Coefficient
- Like terms
- Constant

Additional Terms: N/A

MATERIALS

- Career Profile video
- Algebra tiles (concrete and/or virtual)
- (Optional) Chart paper
- (Optional) Marker

ADDITIONAL PREPARATION

- 9.1.1 Warm-Up contains a video to accompany the question prompts (three-minute video).
- If using concrete algebra tiles, determine how many tiles are needed to model all of the examples in the lesson and organize student sets in a container or plastic bag.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may try to combine terms that are not like terms, for example, $2 + 3x = 5x$ or $3x + 2y = 5xy$.
- Students may not recognize the coefficient of 1 when adding like terms, for example, $x + 2x + 3x = 5x$ rather than $6x$.

THINGS TO CONSIDER

- Determine if your students need the support of all or part of the refresher.
- Consider making student anchor charts, Frayer Models, or a word wall to reinforce the vocabulary terms presented in 9.1.3 Guided Instruction and 9.1.4 Exploration that will be referenced throughout the unit.
- Notice the intentional spiraling throughout the next four lessons. Students do not need mastery yet, but instead have opportunities to continue practicing adding and subtracting over the next three lessons.
- If time is an issue, then consider assigning 9.1.7 Your Turn! for homework.

LITERACY AND LANGUAGE ROUTINES

Notice and Wonder

Make use of this routine in 9.1.2 Refresher in order to position students to make sense of the rationale behind why variables like x and y are not considered like terms. The visual representation of algebra tiles enhances this routine.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.2.1 Multiple Representations

Multiple portions of this lesson provide opportunities for students to represent algebraic expressions in various ways. In 9.1.2 Refresher, students have the opportunity to see multiple ways to represent numeric expressions both using algebra tiles and creating other equivalent expressions.

DIFFERENTIATE INSTRUCTION

Consider providing algebra tiles, having students sketch algebra tiles, or providing access to the digital version of these manipulatives as an alternative method to capturing how students are processing the meaning of zero pairs and simplifying expressions.

9.1.1 WARM-UP: CAREER IN ROBOTICS

QUESTIONING

Do you know someone who is an engineer or computer programmer? Consider bringing in a visitor for your class to ask questions and learn from. Additionally, consider having students return to this lesson later in the year for project-based learning around this Career Profile.

TEACHER MOVES

Ask students to explain why they have or have not considered the profession. Consider specifying how many examples students should come up with. Is one example okay or should students write more than one example? Is a text-based response sufficient or would you like students to include a diagram (of their robot invention and/or the task it performs?

9.1.1 Warm-Up: Career in Robotics

Tessa Lau is the Chief Robot Whisperer for a robotics company. The company creates robot helpers for the service industry. Some of their robots perform tasks like delivering items to guests at a hotel. Tessa has studied applied physics, computer science, artificial intelligence, and human-computer interaction. She loves hands-on experiments, building things, and the speed of creating with computers. Through programming and engineering, one of her primary jobs is to look at a robot in context and make sure the robots behave well in human environments.

1. Have you ever considered a career as an engineer?
   Answers will vary.

2. How do you think a robot could be helpful to you in your everyday life?
   Answers will vary.

3. Tessa's advice is "Don't listen to someone who tells you that you can't do something." Describe a time someone told you that you were not able to do something.
   Answers will vary.

9.1.2 Refresher: Adding and Subtracting Integers with Algebra Tiles

Algebra tiles, like integer chips, can be used to model adding and subtracting integers. Each small square represents a positive 1 (blue) or a negative 1 (red).

Integer Chips	Algebra Tiles
+ −	1 -1

Adding a positive 1 (blue) and a negative 1 (red) represents a zero pair because $1 + {}^{-}1 = 0$, as shown.

1 -1 $= 0$

1. Describe why this diagram represents the value 0.

1 -1
1 -1
1 -1

Zero pairs result from the sum of each of the three sets of opposites. Algebraically: $1 + (-1) + 1 + (-1) + 1 + (-1) = 0 + 0 + 0 = 0.$

A ***zero pair*** is a pair of numbers whose sum is zero.

2. Complete the following problems.

   a. Complete the table.

Numeric Expression	Algebra Tiles Representation	Value
$-7 + 2$	-1 -1 -1 -1 -1 -1 -1 1 1	-5
$*6 - 8$	1 1 1 1 1 1 1 1 -1 -1	-2
$*-1 - 5$	-1 -1 -1 -1 -1 -1 1 1 1 1 1	-6
$*4 - (-3)$	1 1 1 1 1 1 1 -1 -1 -1	7

   b. For some of the representations with algebra tiles, zero pairs had to be added to the representation. Put an asterisk (*) next to the expressions where this was the case.

   c. Explain why zero pairs had to be added to the representation in these cases.

   For $6 - 8$, two zero pairs were added to be able to subtract positive 8.

   For $-1 - 5$, five zero pairs were added to be able to subtract positive 5.

   For $4 - (-3)$, three zero pairs were added to be able to subtract negative 3.

9.1.2 REFRESHER: ADDING AND SUBTRACTING INTEGERS WITH ALGEBRA TILES

TEACHER MOVES

Choose a student to read the definition of zero pairs out loud and briefly discuss how zero pairs are represented using algebra tiles. Students then complete a table to model numeric expressions with algebra tiles. Before answering question 2b and 2c, consider displaying student work that used zero pairs and student work that did not to elicit a discussion about when utilizing zero pairs is necessary and effective.

- How are the representations of zero pairs with integer chips and algebra tiles the same?
- How are they different?

DIFFERENTIATE INSTRUCTION

Consider the different entry points and ways to engage with key vocabulary, such as zero pair. Word walls, vocabulary cards, and sentence stems are all strategies that can serve multiple learning styles with vocabulary content.

TEACHER MOVES

Students can either place their algebra tiles in each blank table or sketch a depiction of their algebra tiles that represent each expression

!!!.

TEACHER MOVES

Notice and Wonder

The purpose of this activity is to refresh prior learning with adding and subtracting integers before moving into adding and subtracting algebraic expressions.

- For which expressions did zero pairs need to be added to the model to find their value?
- Why was it necessary to add zero pairs in these cases?

9.1.3 GUIDED INSTRUCTION: ALGEBRA VOCABULARY

DIFFERENTIATE INSTRUCTION

Consider a variety of strategies to incorporate this math vocabulary into your classroom.

- Visual Scaffolding: Learning tools like foldables or a Frayer Model provide examples and nonexamples for students to make sense in context.
- Positive Affirmation: Consider using a token economy or form of positive affirmation (snaps) when a student correctly utilizes glossary terms in context when responding.
- Vocab Captain: Nominate a different student per day to monitor opportunities in which students could have replaced an informal word with a math glossary term. Consider giving the class a homework pass for each class period in which there were zero misses.
- Visual Word Wall: Create a visual word wall where students see the vocabulary term and a corresponding example.

9.1.3 Guided Instruction: Algebra Vocabulary

There are some fundamental definitions that are necessary to know for working with algebraic expressions.

A ***term*** is a single number or variable, or numbers and variables multiplied together. For example, the expression $0.5 - 4y$ has two terms.

The expression $0.5 - 4y$ has two terms.

0.5	$-4y$
term 1	term 2

One of the terms in this expression is a constant term.

1. Which term is the constant term? Explain.
   The term 0.5 is the constant because it is not multiplied by a variable.

An ***expression*** is a mathematical statement containing numerals, operators, grouping symbols, and symbols or variables for unknown values. An expression does not contain an equal sign or inequality symbol.

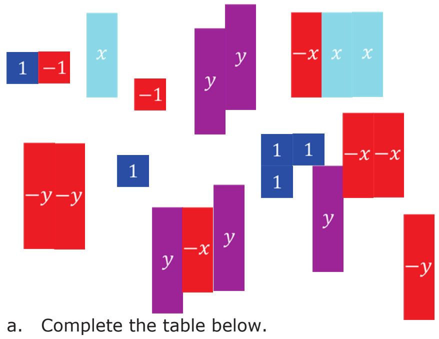
For example, $4 + 5$, $3x - 2$, and $6x - 7y + 11$ are all expressions.

A ***coefficient*** is the number or constant that multiplies a variable in an algebraic expression. If no number is specified, the coefficient is 1.

In the expression $4xy$, 4 is the coefficient.
In the expression x, 1 is the coefficient.
In the expression $-8y$, -8 is the coefficient.
Variable terms have coefficients while constant terms do not.

9.1.4 Exploration: What are Like Terms?

1. Here is a collection of algebra tiles.

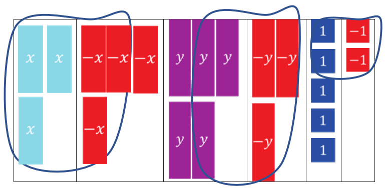


a. Complete the table below.

Type of Algebra Tile	Number of Tiles
x	3
$(-x)$	4
y	5
$(-y)$	3
1	5
(-1)	2

b. Use the definitions of coefficient and term to complete the expression by writing the coefficient or constant term that matches the number of tiles of each type.

$$3x + (-4x) + 5y + (-3y) + 5 + (-2)$$



c. Draw the number of algebra tiles of each type under the term in the expression.

d. Circle the zero pairs in your drawing.

e. Complete the expression below after eliminating the zero pairs.

$$-x + 2y + 3$$

Like terms are terms with the exact same variable(s) raised to the same exponent.

A ***constant*** is a fixed numerical value with no variables. All constants are like terms.

9.1.4 EXPLORATION: WHAT ARE LIKE TERMS?

DIFFERENTIATE INSTRUCTION

Consider providing algebra tiles for students or groups of students as kinesthetic scaffolding.

TEACHER MOVES

Provide space for students to explore and make mistakes as they begin developing their understanding of variables, coefficients, and like terms.

As students explore, monitor for students who might interpret the three light blue algebra tiles as x + 3 instead of 3x. Consider asking those students to describe what x + 3 might look like with tiles and compare that to what they see here. (Notice the answer key in 9.1.6 Guided Practice is a sample of what this could look like.)

9.1.4 EXPLORATION: WHAT ARE LIKE TERMS? (CONTINUED)

TEACHER MOVES

Consider a Turn and Talk for question 2 in which students first answer parts a-c on their own before sharing with a partner to refine their responses.

2. Answer the following questions about like terms.

   a. Classify the following terms.

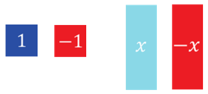
	Like Terms	Unlike Terms
y and 6	○	●
$3x$ and 3	○	●
$5x$ and $-3x$	●	○
10, -2, 7.5	●	○

   b. How did you determine whether the terms were like or unlike?
   Answers will vary. If the terms had the exact same variable(s) raised to the same exponent, they are like terms. If the terms did not, they were unlike.

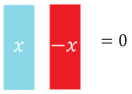
   c. Compare your answers with your partner.

9.1.5 Guided Instruction: Adding and Subtracting Expressions with Algebra Tiles

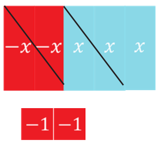
The algebra tiles shown represent x and $-x$ as rectangles, and integers as squares. Red represents the negative of that variable or constant, while the blue shades represent the positive.



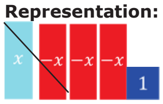
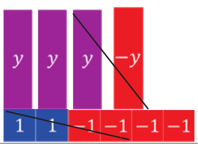
Variables also form zero pairs, as shown.



1. Complete the table by crossing out the zero pairs and recording how many there are. Write the resulting algebraic expression after eliminating the zero pairs.

Algebra Tiles	Number of Zero Pairs	Algebraic Expression
Representation: 	*2 zero pairs of x*	$x - 2$

2. Draw a representation of the following expressions with algebra tiles. Cross out any zero pairs. Then write an equivalent algebraic expression after eliminating any zero pairs.

Algebraic Expression and Representation	Equivalent Algebraic Expression
$x - 3x + 1$ **Representation:** 	$-2x + 1$
$-4 + 3y - y + 2$ **Representation:** 	$2y + (-2)$ *or* $2y - 2$

9.1.5 GUIDED INSTRUCTION: ADDING AND SUBTRACTING EXPRESSIONS WITH ALGEBRA TILES

TEACHER MOVES

For students new to algebra tiles, consider discussing the dimensions of the tiles, noting that "like" terms have "like" sizes. This would be important to note for any students sketching tiles.

DIFFERENTIATE INSTRUCTION

Consider providing algebra tiles for students or groups of students as kinesthetic scaffolding.

TEACHER MOVES

As students transition from numeric expressions to algebraic expressions, it becomes more essential to highlight the differences in size as an indicator for the operations. Transition student language from "same size" to "like terms" for effective transitions to algebraic expressions.

9.1.6 GUIDED PRACTICE: ADDING AND SUBTRACTING EXPRESSIONS WITH ALGEBRA TILES

ENRICHMENT

Consider a Turn and Talk or Think-Pair-Share for students or classrooms who learn best with a partner.

- How are these expressions the same? How are they different?
- If the two expressions were added together, what would be the resulting expression?

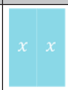
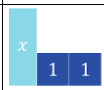
QUESTIONING

Consider challenging students to provide mathematical justification by substituting a value for x to prove they are not equivalent.

9.1.6 Guided Practice: Adding and Subtracting Expressions with Algebra Tiles

1. Consider the expressions $2x$ and $x + 2$.

   a. Draw a model of $2x$ and $x + 2$ with algebra tiles.

Expression	Algebra Tiles Model
$2x$	
$x + 2$	

   b. How are they different?

   The expression $2x$ represents the sum of two like terms: $x + x$. Equivalently, it represents the product: $2x$. On the other hand, $x + 2$ represents the sum of two terms that are not like terms, x and 2, so they cannot be combined.

   c. Are $2x$ and $x + 2$ equivalent algebraic expressions?

   No, see part b for justification.

   $x + x \neq x + 2$

9.1.7: YOUR TURN!

DIFFERENTIATE INSTRUCTION

Consider giving students the opportunity to use or sketch algebra tiles or use the digital manipulatives as a method of producing their answers.

9.1.7 Your Turn!

Write an equivalent expression by combining like terms.

1. $2x + 4x =$ $6x$
2. $-y + 7 + 4y - 1 =$ $3y + 6$
3. $x - x + x - x + 1 =$ 1
4. $8x - 4 - 4x =$ $4x - 4$
5. $2x - 4x =$ $-2x$

9.1.8 Wrap-Up: Cool Down

1. For each key word, check the box in the column that describes your knowledge of the word. Then answer the questions or complete the statements in the last column.
Answers will vary.

Key Word	I already knew the meaning of this word.	I've heard this word before, but I was not sure what it meant.	Questions
Expression	☐	☐	An example of a numeric expression is ___3___. An example of an algebraic expression is ___$2x+4$___.
Term	☐	☐	In the expression $8x - 9 + 2y$, how many terms are there? ___3___
Coefficient	☐	☐	In the expression $8x - 9 + 2y$, what are the coefficients? ___8 and 2___
Equivalent	☐	☐	When working with algebraic expressions the word "equivalent" means ___*two expressions have the same value when the variables are set to the same numbers*___.

9.1.8 WRAP-UP: COOL DOWN

DIFFERENTIATE INSTRUCTION

Consider utilizing students' responses to inform how to scaffold instruction for Lesson 2 when adding linear expressions.

TEACHER MOVES

If students state that -9 is a coefficient, further ask them why. Theoretically it is a coefficient of x^0. If a student does not use that reasoning, then the student has the misconception that any numeric value is a coefficient.

